

JOSEPH MWANGI NDUNG’U

Computer Science Graduate | Software Developer | AI & Machine Learning | Data & Field Research
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PROFILE

Computer Science graduate from Machakos University, specializing in Artificial Intelligence and Machine Learning, with strong practical experience in full-stack software development, data engineering, field research, and enterprise ICT systems. Demonstrated ability to design, build, and deploy end-to-end technical solutions - from web applications and RESTful APIs to automated data pipelines, networked systems, and field-based data collection projects involving direct community engagement. Complemented by real-world exposure to large-scale ICT infrastructure through an internship at Kenya Power & Lighting Co. (KPLC). A fast learner, effective communicator, and collaborative team player who is equally comfortable working independently in the field or within cross-functional teams.

TECHNICAL SKILLS

Programming	Python, JavaScript, Java, SQL, HTML/CSS — applied across web apps, automation scripts, and data pipelines
Web Development	Django, RESTful APIs, full-stack application design and deployment; responsive front-end development
AI & Machine Learning	TensorFlow, Keras, Scikit-learn, Pandas, NumPy; model design, training, evaluation, and deployment
Data & Databases	SQL database design, data cleaning, validation, pipeline construction, reporting and visualisation (Power BI)
Field Research & Data Collection	On-site field data collection, survey administration, KoboToolbox-style digital data capture, stakeholder interviews and engagement
Networking & ICT	Network infrastructure, hardware diagnostics, SCADA systems, structured cabling fundamentals, TCP/IP
Soft Skills	Stakeholder communication, technical documentation, cross-functional teamwork, analytical problem-solving, professionalism in community-facing settings

PROJECTS

Plant Disease Detection System | Independent Project | Completed & Deployed (2024-2025)

- Designed and deployed an AI-powered image classification system using Convolutional Neural Networks (CNNs) in Python (TensorFlow, Keras) to identify plant diseases from photographs with high accuracy.
- Conducted field research across multiple farms, visiting and interacting directly with farmers to understand how they identified, managed, and responded to crop diseases - using these insights to inform what the system needed to detect and how results should be communicated to end users.
- Administered structured questions and informal interviews with farmers to capture real-world disease symptoms, local terminology, and existing coping practices, then translated these findings into the dataset categories and labeling framework used to train the model.
- Built an end-to-end data pipeline: image data collection (including field-sourced samples), preprocessing, augmentation, model training, iterative evaluation, and deployment - managing over 50,000 image samples across multiple disease classes.
- Developed a full-stack Django web application with a RESTful API, enabling non-technical users (including farmers) to upload images and receive instant, data-driven disease diagnoses.
- Implemented a structured SQL database for storing predictions, user records, and system logs; built a reporting interface for tracking model performance metrics over time.
- Produced comprehensive technical documentation and user guides to support adoption and long-term maintenance of the system.

PROFESSIONAL EXPERIENCE

ICT Intern - Telecommunication Department | *Kenya Power & Lighting Co. PLC (KPLC), Nakuru* |
May 2025 – Jul 2025

- Embedded within the ICT division of Kenya's largest power utility, gaining direct exposure to enterprise-scale network infrastructure, hardware systems, and operational ICT workflows across a large, multi-site organization.
- Diagnosed and resolved complex technical faults across hardware, software, networking, and SCADA monitoring systems - applying structured troubleshooting methodologies in a mission-critical environment.
- Supported SCADA system operations, developing hands-on familiarity with real-time, data-driven industrial monitoring and control infrastructure.
- Maintained enterprise ICT inventory and coordinated hardware deployments across departments, applying process documentation and asset management best practices.
- Engaged daily with department heads, external suppliers, and end users to resolve issues and communicate technical findings clearly to non-technical stakeholders.

EDUCATION

BSc Computer Science - AI & Machine Learning Specialization | *Machakos University, Kenya* |
2021 - 2026 | Completed May 2026, Graduating November 2026

- Relevant coursework: Machine Learning, Deep Learning, Computer Networks, Database Systems, Software Engineering, Cybersecurity, Computer Vision, Algorithms & Data Structures, Statistics & Probability, Operating Systems.
- Capstone project (Plant Disease Detection System) demonstrates ability to manage a complex, multi-phase technical system from initial concept and field research through iterative development to live deployment.

ADDITIONAL INFORMATION

- Languages: English (fluent), Swahili (fluent)
- Comfortable conducting field work, community engagement, and interacting professionally with diverse stakeholders
- Interests: Open-source development, AI for social good, networking technology, community tech initiatives
- Available for internship, attachment, or entry-level roles - immediate availability